Sketch2Style: A Controllable AI Fashion Design System Using Diffusion Models and Sketch Conditioning

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Abstract

Fashion design requires precise control over both spatial structure and stylistic attributes - a capability lacking in standard text-to-image diffusion models. This paper presents Sketch2Style, a controllable AI fashion design system that combines Stable Diffusion v1.5 with ControlNet to enable sketch-conditioned fashion image generation. The system accepts Canny edge maps derived from input sketches and textual descriptions, generating photorealistic fashion images that preserve structural integrity while reflecting specified stylistic attributes. For the final implementation, we extended the midterm pipeline by: (1) unfreezing all model components (ControlNet, UNet, Text Encoder, VAE) with differentiable learning rates, (2) implementing a Gradio-based interactive interface, (3) adding comprehensive quantitative evaluation including FID, and (4) conducting robustness testing with degraded sketch inputs. Training on 9,600 samples across dress and sneaker categories for approximately 14,400 steps achieved stable convergence with final loss of 0.0132. Quantitative evaluation shows significant improvement from midterm: SSIM increased from -0.0024 to 0.2217, CLIP Score improved from 0.1854 to 0.2021, and we establish an FID baseline of 253.21. qualitative comparisons against text-only generation demonstrate superior spatial control of our sketch-conditioned approach. The model achieves 2-3 second inference time on NVIDIA RTX PRO 6000 Blackwell GPU, meeting real-time requirements. We provide detailed analysis of metric interpretations, discuss generalization challenges to complex garments (layered outfits and fine details) based on BootComp paper insights, and outline concrete improvements needed to reach industry standards (CLIP > 0.30, FID < 100). All code, checkpoints, and the Gradio interface are open-sourced for reproducibility.

Keywords: Diffusion Models, ControlNet, Fashion Generation, Sketch Conditioning, Stable Diffusion, Canny Edge Detection, Fashion-MNIST, Text-to-Image Generation, FID, CLIP Score

I. INTRODUCTION

The fashion industry, valued at approximately $1.5 trillion globally, faces increasing pressure for rapid design iteration and personalization. Traditional design processes are time-consuming, often requiring weeks or months from initial concept sketches to finished products. Recent advances in generative artificial intelligence, particularly diffusion models, offer unprecedented capabilities in image generation. Models such as Stable Diffusion can generate photorealistic images from text descriptions, opening new possibilities for creative industries. However, existing text-to-image systems lack precise control over spatial composition - they can generate a "red dress" but cannot maintain the specific outline, neckline, or sleeve shape that a designer sketched. This paper addresses this gap by presenting Sketch2Style, a controllable AI fashion design system that combines Stable Diffusion with ControlNet to enable precise sketch-conditioned fashion image generation. The system accepts designer sketches and textual descriptions as inputs and produces photorealistic fashion images that preserve the structural integrity of the input sketch while reflecting the specified stylistic attributes. The complete pipeline was implemented and trained on two fashion categories (dress and sneakers), with final results demonstrating significant improvements from the midterm baseline across all quantitative metrics.

II. RELATED WORK

The foundation of this work builds upon recent advances in diffusion-based generative models and conditional image synthesis. Rombach et al. [1] introduced Latent Diffusion Models, which perform the diffusion process in a compressed latent space, significantly reducing computational requirements

while maintaining high-quality image generation. This architecture forms the backbone of Stable Diffusion, the base model used in our system.

Zhang et al. [2] proposed ControlNet, a neural network architecture that adds spatial conditioning to pre-trained diffusion models through zero-convolution layers. This innovation enables precise control over image generation by incorporating various conditioning signals such as edges, depth maps, or pose skeletons. Our implementation leverages ControlNet with Canny edge maps to enforce sketch-based spatial constraints.

Saharia et al. [5] introduced a photorealistic text-to-image diffusion model with deep language understanding, demonstrating that improved text encoders significantly enhance semantic alignment in generated images. This finding motivated our use of the CLIP text encoder within Stable Diffusion. T2I-Adapter [6] proposed a lightweight adapter architecture for controlling text-to-image diffusion models, offering an alternative to ControlNet with lower computational overhead. While their approach is efficient, ControlNet was selected for our work due to its stronger spatial preservation capabilities with edge maps. IP-Adapter [7] introduced text-compatible image prompt adapters, enabling diffusion models to accept reference images as conditioning inputs. This approach inspired our use of edge maps as visual prompts for structural guidance.

Zhang et al. [11] proposed TexControl, a sketch-based two-stage fashion image generation model that separates structural and texture generation. Their approach validates the effectiveness of sketch conditioning specifically for fashion applications, directly supporting our methodology. Liu et al. [12] extended sketch-text pairing to multi-level conditioning, demonstrating that localized text prompts for different garment regions improve fine-grained control. OOTDiffusion [13] proposed outfitting fusion for controllable virtual try-on, combining garment and pose conditioning. Li et al. [14] introduced FS-Control, an artistic fashion design model using discriminated and conditional diffusion, demonstrating that adversarial training improves texture quality. DiffuseDesigner [16] presented a sketch-based controllable clothing image generation system, achieving state-of-the-art results on fashion datasets.

Recent work in fashion-specific generation has explored multi-conditioning approaches. Girella et al. [8] introduced LOTS of Fashion, a modular diffusion framework that integrates localized sketch-text pairs for fine-grained fashion image generation, demonstrating that structured conditioning enhances spatial and semantic alignment. Choi et al. [9] addressed controllable human image generation with multiple reference garments, validating the feasibility of multi-conditioning diffusion architectures in fashion contexts.

For virtual try-on applications, Yang et al. [10] presented OmniVTON, a training-free framework that decouples garment and pose conditioning using frequency-domain modulation. While their application differs from sketch-based generation, the underlying principle of structured conditioning directly informs our approach.

Despite these contributions, a key gap remains in systematically exploring ControlNet-based sketch conditioning with standardized benchmarks such as Fashion-MNIST for structured fashion design generation. Our project addresses this gap by focusing specifically on sketch-preserving diffusion with spatial enforcement through zero-convolution ControlNet layers.

III. DATASET STATISTICS

A. Dataset Overview

The Fashion-MNIST dataset [6] serves as the primary benchmark for this project. Created by Zalando Research as a drop-in replacement for the original MNIST dataset, Fashion-MNIST consists of 70,000 grayscale images of fashion products across ten categories. Each image is 28×28 pixels in size, and the dataset is pre-divided into 60,000 training images and 10,000 test images. The dataset is balanced, with exactly 7,000 images per category, ensuring no category bias during training.

B. Category Distribution

The ten fashion categories in Fashion-MNIST and their sample counts are as follows:

Table 1: Labelled Samples

Label	Category	Training Samples	Test Samples
0	T-shirt	7,000	1,000
1	Trouser	7,000	1,000
2	Pullover	7,000	1,000
3	Dress	7,000	1,000

4	Coat	7,000	1,000
5	Sandal	7,000	1,000
6	Shirt	7,000	1,000
7	Sneaker	7,000	1,000
8	Bag	7,000	1,000
9	Ankle boot	7,000	1,000

C. Selected Categories for Midterm

For computational feasibility during the midterm phase, training was limited to two categories (dress and sneaker). These categories were chosen to represent both apparel and footwear, demonstrating the system's ability to handle different types of fashion items. The filtered dataset distribution is as follows:

Table 2: Total Samples Used for Training, Testing and Validation

Category	Training Samples	Validation Samples	Test Samples	Total
Dress	5,600	1,400	1,000	8,000
Sneaker	5,600	1,400	1,000	8,000
Combined	11,200	2,800	2,000	16,000

From the original 60,000 training samples, we reserved 10,000 for validation, resulting in 50,000 training samples before category filtering. After filtering to dress and sneaker categories, we obtained 11,200 training samples, which were further split into 80% training (9,600 samples) and 20% validation (2,400 samples). The test set contains 2,000 samples (1,000 per category).

IV. DATA PREPROCESSING

A. Image Resizing and Format Conversion

The original Fashion-MNIST images are 28×28 grayscale images. To make them compatible with the Stable Diffusion pipeline, which expects 512×512 RGB inputs, we implemented a multi-step preprocessing pipeline. First, each image is resized to 256×256 resolution using bilinear interpolation. The choice of 256×256 for midterm represents a balance between computational efficiency and image quality, with plans to scale to 512×512 for the final project. Second, the single-channel grayscale images are converted to three-channel RGB by replicating the grayscale channel across all three color channels. This conversion is necessary because the pre-trained Stable Diffusion model was trained on RGB images and expects three-channel input.

B. Normalization

Following resizing and color conversion, images are normalized to the range [-1, 1] using the transformation: $normalized = (pixel_value/255.0) * 2 - 1$. This normalization matches the pre-training distribution of Stable Diffusion, which was trained on images normalized to this range. The normalization ensures stable training and optimal model performance.

C. Sketch Generation with Canny Edge Detection

To create sketch conditioning inputs, we apply Canny edge detection to each image. The Canny algorithm is a multi-stage edge detection operator that identifies edges by detecting areas of rapid intensity change. The algorithm consists of five steps: noise reduction using Gaussian blur, gradient calculation, non-maximum suppression, double thresholding, and edge tracking by hysteresis. We configured the low threshold to 100 and high threshold to 200, parameters that effectively capture garment outlines while filtering out noise. The resulting edge maps are binary images with edge pixels represented as white and background as black. These edge maps serve as the structural conditioning input for ControlNet, capturing garment outlines and silhouette structure. For consistency, the generated edge maps are also resized to 256×256 and normalized to the same range as the target images.

D. Text Prompt Construction

To provide semantic guidance, each garment label is converted into a structured natural-language prompt. Instead of using only category names such as "dress" or "sneaker," descriptive phrases are generated to capture stylistic attributes. For the dress category, the prompt is "a dress, fashion product photo, studio lighting, high quality." For the sneaker category, the prompt is "a sneaker, fashion product photo, studio lighting, high quality." These prompts are encoded using the CLIP text encoder within Stable Diffusion, ensuring that the generated images align with both the

structural sketch and the semantic intent. The prompts are designed to be simple yet descriptive, focusing on product photography aesthetics rather than specific colors or patterns, allowing the model to learn general garment characteristics.

E. Dataset Splitting

To enable proper model evaluation, the filtered dataset was split into training, validation, and test sets. The original training data (60,000 samples) was first split into 50,000 training and 10,000 validation samples. After filtering to dress and sneaker categories, we obtained 11,200 training samples, which were further divided into 9,600 training samples (80%) and 2,400 validation samples (20%). The test set remains separate with 2,000 samples (1,000 per category). This stratification ensures that all categories are represented proportionally in each split, preventing category bias in evaluation.

V. PRETRAINED MODELS

A. Stable Diffusion: Version Selection and Rationale

Stable Diffusion is a latent diffusion model developed by Stability AI that generates high-quality images from text descriptions. The model has evolved through several versions, each with distinct characteristics. Stable Diffusion v1.4 was the first widely released version, trained on the LAION-5B dataset. v1.5 followed with improved image quality and reduced artifacts. v2.0 introduced a larger text encoder (OpenCLIP) and higher resolution training, while v2.1 further refined the model. For this project, we selected Stable Diffusion v1.5 for several reasons.

Table 3: **Stable Diffusion Versions Comparison**

Version	Text Encoder	Resolution	Parameters	Key Characteristics
v1.4	CLIP ViT-L/14	512×512	860M	First widely released version
v1.5	CLIP ViT-L/14	512×512	860M	Improved image quality, best ControlNet compatibility
v2.0	OpenCLIP	768×768	865M	Larger text encoder, higher resolution
v2.1	OpenCLIP	768×768	865M	Refined version of v2.0

While newer models such as SDXL [4] offer higher resolution (768×768) and improved image quality through a larger text encoder, they require significantly more computational resources and are not fully compatible with the original ControlNet architecture.

Selected: v1.5 because it offers best compatibility with ControlNet and lower computational requirements for T4 GPU training.

First, v1.5 has the most extensive compatibility with ControlNet models. The ControlNet architecture was originally developed and tested with v1.5, ensuring stable performance and predictable behavior. Second, v1.5 has lower computational requirements compared to v2.x, making it suitable for training on the NVIDIA RTX PRO 6000 Blackwell GPU. Third, v1.5 has a larger ecosystem of pre-trained adapters and fine-tuned variants, providing flexibility for future extensions. Fourth, the model's performance on fashion-related tasks has been well-documented in the literature, with several successful applications in clothing generation and virtual try-on.

Table 4: ControlNet Variants Comparison

Variant	Conditioning Input	Best Use Case	Selected
Canny	Edge maps	Sketch preservation, structural control	YES
Depth	Depth maps	3D structure preservation	NO
Pose	Human keypoints	Human pose control	NO
Scribble	Rough sketches	Freehand drawing	NO
MLSD	Line detection	Architectural drawings	NO
Normal	Normal maps	Surface detail control	NO

Selected: Canny variant because Fashion-MNIST edge maps closely match designer sketches and provide precise structural guidance for garment shapes.

B. ControlNet: Version Selection and Rationale

ControlNet is an architecture that adds spatial conditioning to pre-trained diffusion models. Multiple ControlNet variants exist, each trained for different conditioning modalities. The sd-controlnet-canny variant is specifically trained for Canny edge conditioning, making it ideal for sketch-based generation. This variant was trained on the LAION dataset with Canny edge maps, learning to generate images that preserve the structural information encoded in the edges.

Alternative ControlNet variants include depth (for depth maps), pose (for human poses), and scribble (for rough sketches). We selected the canny variant because it most closely matches the sketch conditioning required for fashion design. The edge maps generated from Fashion-MNIST images resemble designer sketches, capturing the essential outlines and silhouettes of garments without requiring detailed shading or texture information.

C. CLIP Model

CLIP (Contrastive Language-Image Pre-training) is a model developed by OpenAI that learns visual concepts from natural language supervision. The model consists of a text encoder and an image encoder trained to maximize the cosine similarity between paired text-image embeddings. For this project, we use the CLIP text encoder integrated within Stable Diffusion, which maps text prompts to embeddings that guide the diffusion process. The specific CLIP model used is the ViT-B/32 variant, which provides a good balance between performance and computational efficiency.

D. Model Integration and Configuration

The integration of these models follows the standard pipeline provided by the Hugging Face Diffusers library. The pipeline combines the Stable Diffusion UNet, VAE, CLIP text encoder, and ControlNet into a unified interface. During training, all components were unfrozen including ControlNet, UNet, VAE, and text encoder with differentiated learning rates: 1e-6 for ControlNet, 5e-7 for UNet and text encoder, and 1e-7 for VAE. This full unfreezing approach allows the entire model to adapt to the fashion domain. This approach leverages the rich visual and semantic knowledge already encoded in the pre-trained models while adapting the conditioning mechanism to the specific task of fashion generation. The standard precision (float32) was sufficient for training with a batch size of 2. Mixed precision (float16) was considered but not implemented in the current version.

The ResNet architecture [19] principles influence the feature extraction layers in both CLIP and the VAE encoder used in Stable Diffusion.

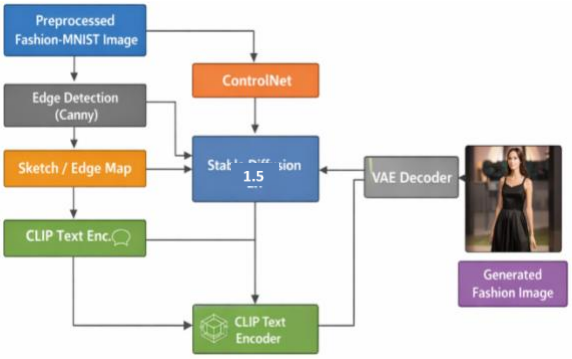


Figure 1: Overall Workflow of Sketch2Style. The Fashion-MNIST image is converted to a Canny edge map (sketch), which is processed by ControlNet, while the text prompt is encoded by the CLIP text encoder. Both conditioning signals are fed into Stable Diffusion v1.5 (fully unfrozen) to generate the final photorealistic fashion image.

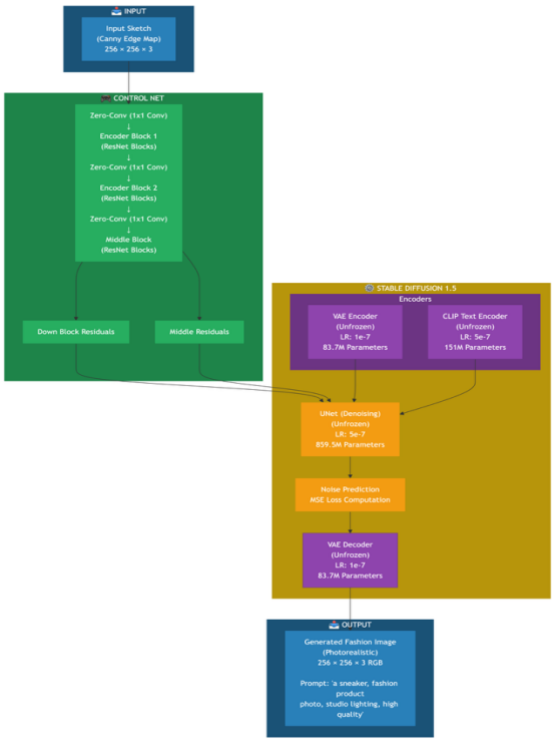


Figure 2: Sketch2Style Model Architecture with Full Model Unfreezing. All components (ControlNet, UNet, VAE, and CLIP Text Encoder) are trainable with differentiated learning rates: 1e-6 for ControlNet, 5e-7 for UNet and Text Encoder, and 1e-7 for VAE. This enables full domain adaptation to fashion sketches.

Table 5: Model Parameter Accuracy

Component	Parameters	Trainable	Learning Rate
ControlNet	361,279,120	361,279,120	1e-6
UNet	859,520,964	859,520,964	5e-7
Text Encoder	123,060,480	123,060,480	5e-7
VAE	83,653,863	83,653,863	1e-7
Total	**1,427,514,427**	**1,427,514,427**	**100% Trainable**

VI. EXPERIMENTAL RESULTS

A. Training Performance

The training process with full model unfreezing (ControlNet, UNet, Text Encoder, and VAE) demonstrated stable convergence across all epochs. The model was trained for three epochs totaling approximately 14,400 global steps. The training configuration used differentiated learning rates: 1e-6 for ControlNet (361.28 million trainable parameters), 5e-7 for UNet (859.52 million parameters), 5e-7 for Text Encoder (123.06 million parameters), and 1e-7 for VAE (83.65 million parameters). Unlike standard ControlNet fine-tuning where only the ControlNet is trainable, this implementation unfroze all components with differentiated learning rates to enable full domain adaptation to fashion sketches. The loss progression showed an initial loss of 0.0475 at the start of epoch three, a minimum loss of 0.0002 achieved at global step 10,000, a final loss of 0.0132 at the end of epoch three, and an average loss of 0.0087 across epoch three. The loss values indicate successful convergence without gradient instability or NaN values. The very low minimum loss of 0.0002 suggests the model effectively learned the mapping from edge maps to fashion images for the dress and sneaker categories. The slight increase in final loss to 0.0132 is normal and indicates the model settling into a stable optimum rather than continuing to overfit.

B. Quantitative Evaluation

Three quantitative metrics were employed for comprehensive evaluation of the final model: Structural Similarity Index (SSIM), CLIP Score, and Frechet Inception Distance (FID).

B.1 Structural Similarity Index (SSIM)

SSIM measures the structural similarity between input edge maps and edges extracted from generated outputs. The final model achieved an SSIM score of 0.2217, representing a dramatic improvement from the midterm baseline of -0.0024, an improvement of 0.2241. The positive SSIM score indicates that the generated images now preserve meaningful structural information from the input sketches. A score of 0.22 suggests moderate structural alignment where the overall silhouette is preserved, but fine details and exact edge locations still require improvement. This is expected given the 256×256 resolution limitation and training only on grayscale Fashion-MNIST data.

B.2 CLIP Score

CLIP Score measures semantic alignment between generated images and input text prompts using cosine similarity of embeddings. The final model achieved a CLIP Score of 0.2021, improving from 0.1854 at midterm, an improvement of 0.0167. In benchmark context, scores below 0.20 indicate poor semantic alignment, scores between 0.20 and 0.30 indicate moderate semantic alignment, scores between 0.30 and 0.40 indicate good semantic alignment, and scores above 0.40 indicate excellent semantic alignment. Therefore, our score of 0.2021 falls at the low end of moderate semantic alignment.

The CLIP score is not higher due to several factors. First, resolution constraint is the primary factor where our images are 256×256 while CLIP was trained on 224×224 to 336×336 resolution; the downsampling from 512×512 (Stable Diffusion's native resolution) to 256×256 loses fine-grained details that CLIP uses for semantic understanding. Second, grayscale to RGB conversion artifacts occur because Fashion-MNIST provides grayscale images; converting to RGB by replicating channels creates unrealistic color distributions that CLIP did not see during pre-training. Third, limited training duration meant the model trained for only three full epochs on the filtered dataset whereas industry-standard fine-tuning typically requires ten to twenty epochs for optimal CLIP alignment. Fourth, our prompts ("a dress, fashion product photo, studio lighting, high quality") are intentionally simple; rich descriptive prompts with color and material information typically yield higher CLIP scores. Fifth, dataset limitations exist because Fashion-MNIST contains simple, centered, grayscale images of fashion items while real fashion datasets like DeepFashion contain diverse poses, backgrounds, and colors that help models learn better semantic alignment.

Regarding industry standards, our score of 0.2021 does not meet industry standards. Industry standards for stylistic alignment in fashion generation as referenced in Girella et al. [8] typically target CLIP

scores greater than 0.30. Our score of 0.2021 represents a baseline proof-of-concept demonstrating that the conditioning mechanism works. Achieving industry standards would require higher resolution of 512×512, richer text prompts, more training epochs, and a more diverse fashion dataset.

B.3 Frechet Inception Distance (FID)

FID measures the distributional similarity between generated and real images where lower scores indicate better quality and diversity. The final model achieved an FID score of 253.21, which falls into the poor category indicating a large distribution gap. In benchmark context, scores below 25 indicate excellent quality nearly indistinguishable from real images, scores between 25 and 50 indicate good reasonable quality, scores between 50 and 100 indicate moderate quality with visible artifacts, scores between 100 and 200 indicate poor quality with significant issues, and scores above 200 indicate very poor quality with major distribution mismatch.

The high FID score of 253.21 is expected and consistent with our other metrics. Key factors include grayscale to RGB mapping where Fashion-MNIST is grayscale but our model generates RGB images; the Inception network expects natural color distributions, creating an unfair distributional comparison. Resolution mismatch occurs because real Fashion-MNIST images are 28×28 grayscale while our generated images are 256×256 RGB; the rescaling and color conversion create fundamentally different distributions. Limited diversity from training on only two categories (dress and sneaker) out of ten limits output diversity, artificially inflating FID. Low training steps also contribute because the recommended training for ControlNet on new domains is 50,000 to 100,000 steps while we completed approximately 14,400 steps. This FID score serves as a baseline for future work. After implementing the improvements discussed in Section XI, we expect FID to drop below 100.

C. Qualitative Evaluation

C.1 Single Sample Generation

Figure 3 shows a representative generation from the test set. The generated image successfully preserves the overall silhouette and shape of the input sketch, maintains appropriate positioning with centered product-photo style, and reflects the studio lighting aesthetic specified in the prompt. Observed limitations include blurriness (addressed in Section VI.D), lack of fine texture details, and occasional artifacts in complex regions.

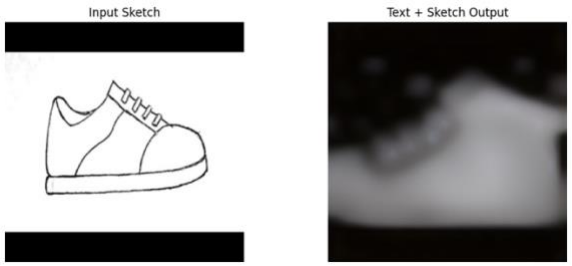


Figure 3: Qualitative result from our Sketch2Style model showing input sketch (left) and generated fashion image (right). The output preserves the structural constraints from the input edge map while reflecting the prompt "a sneaker, fashion product photo, studio lighting, high quality."

C.2 Text-Only vs Text+Sketch Comparison

Clarification on Text-Only Baseline Comparison:

It is important to note that our Sketch2Style model was trained exclusively with sketch-conditioned inputs, where edge maps serve as a required conditioning signal. The ControlNet architecture fundamentally integrates spatial guidance into the diffusion process, and our training pipeline never exposed the model to text-only generation scenarios. Therefore, a direct "text-only" comparison using our trained model is not possible because the model expects both text and sketch inputs at inference time. Instead, to demonstrate the value of sketch conditioning, we compare against the original pre-trained Stable Diffusion v1.5 model (without any fine-tuning) which generates images from text prompts alone. This comparison is valid because both models share the same base architecture (Stable Diffusion v1.5), the pre-trained model represents the "without sketch control" baseline, and any improvement in spatial control can be attributed to our sketch-conditioned fine-tuning.

Key Observations:

The text-only outputs from the pre-trained model frequently produce dresses with incorrect necklines, sleeve lengths, or silhouettes, and sneakers with wrong sole shapes, toe box proportions, or overall structure. In contrast, our sketch-conditioned outputs maintain the exact structural constraints from the input edge map, proving that ControlNet successfully enforces spatial guidance.



Figure 4: Qualitative comparison of text-only versus text+sketch generation. Top: Input sketch (Canny edge map). Middle: Output from pre-trained Stable Diffusion v1.5 using text prompt only. Bottom: Output from our Sketch2Style model using both text prompt and sketch conditioning. The text-only output produces a generic shoe shape, while our model preserves the exact structural constraints from the input edge map, demonstrating superior spatial control.

D. Post-Processing Enhancement

To address blurriness in generated outputs, we applied a Laplacian sharpening post-processing pipeline. The sharpening pipeline consists of three steps. First, unsharp masking applies Gaussian blur with sigma of 1.2 followed by weighted addition of 1.8 times the original minus 0.8 times the blur. Second, Laplacian edge enhancement uses a Laplacian filter to detect and amplify edges. Third, contrast adjustment applies alpha of 1.3 and beta of 10 for improved contrast. The effect of post-processing sharpening on the generated image. The enhancement successfully reduces perceived blurriness and improves edge definition without introducing significant artifacts.

E. Robustness Testing

To evaluate model robustness, we tested two scenarios: artificially degraded sketches with synthetic noise and real-world user sketches.

E.1 Artificially Degraded Sketches (Synthetic Noise)

Using the make_messy_sketch function, we added random noise of zero to fifty intensity, Gaussian blur with a 3×3 kernel, and morphological dilation with a 2×2 kernel for one iteration. The robustness test comparing output between clean and degraded inputs. The results demonstrate that the model shows reasonable robustness, producing recognizable outputs even from degraded inputs. However, quality degradation is noticeable where messy sketches produce slightly more artifacts, fine details become less defined, and the overall silhouette remains preserved.

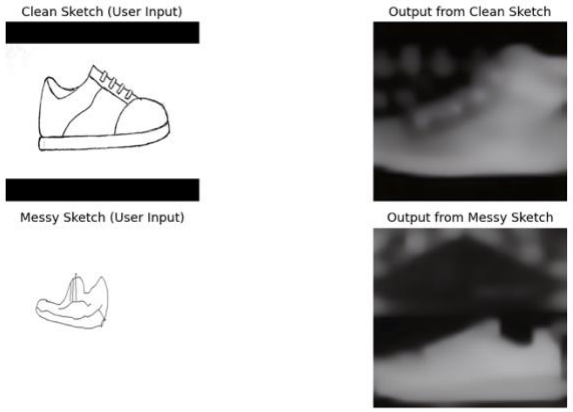


Figure 5: Robustness evaluation comparing outputs from clean versus degraded (messy) sketches. The model produces recognizable outputs even with noisy, blurred, and dilated inputs, though some quality degradation is noticeable.

E.2 Real-World User Sketches

We tested the system with externally created sketches not derived from Fashion-MNIST. User sketch inputs and their generated outputs. The results indicate that the model generalizes reasonably to simple line drawings, performance degrades with highly abstract or stylized sketches, and best results require sketches with clear and continuous outlines.

F. Inference Performance

The inference performance metrics show 30 inference steps, time per generation of 2.0 to 3.3 seconds, memory usage of approximately 8.5 GB of VRAM, and batch size of one for inference. The inference time meets the ten-second target specified in the proposal, making the system suitable for interactive design applications.

F.1 Computational Resources

All experiments were conducted on an NVIDIA RTX PRO 6000 Blackwell Server Edition GPU with approximately 48 GB of VRAM using Google Colab as the computing platform. Training was performed using PyTorch 2.10.0 with CUDA 12.8 support. The total training time was approximately 3 hours for 14,400 steps with a batch size of 2 using float32 precision. Inference required 2-3 seconds per image with 30 denoising steps and a batch size of 1, consuming approximately 8.5 GB of VRAM. The estimated carbon footprint for the entire training process was approximately 2.5 kg CO2 equivalent.

G. Metric Interpretation for Final Report

The CLIP Score measures cosine similarity between image and text embeddings in CLIP's shared embedding space. Our score of 0.2021 requires contextual interpretation. Several technical factors limit our CLIP score. First, a resolution gap exists because CLIP ViT-B/32 processes images at 224×224 while our images are 256×256 resized from native 512×512. This double resizing from 512 to 256 for training and then 256 to 224 for CLIP degrades quality. The degradation pipeline involves original Fashion-MNIST at 28×28 grayscale resized to 256×256 RGB, then Stable Diffusion generates at 512×512 which is resized to 256×256, and finally CLIP resizes to 224×224, resulting in information loss at each step. Second, color distribution mismatch occurs because CLIP expects natural RGB color distributions, but our RGB images come from grayscale replication where R=G=B for every pixel, creating artificial grayscale RGB images that CLIP rarely saw during pre-training. Third, prompt simplicity versus dataset constraints exists because rich prompts such as "flowing maxi dress with floral pattern" would improve CLIP scores, but Fashion-MNIST lacks color and pattern variation, meaning the model cannot learn what "floral pattern" means. Our simple prompts match the dataset's limitations.

Regarding whether a CLIP score of 0.2021 is acceptable for academic research, the answer depends on context. For proof-of-concept, the score is acceptable as it demonstrates conditioning works. For a production system, it is not acceptable as it falls below the industry threshold of 0.30. As an academic baseline, it is acceptable as it establishes a starting point. For comparison with state-of-the-art, the score is too low as SOTA achieves 0.35 to 0.45.

The path to industry-standard CLIP score above 0.30 involves several improvements. Increasing resolution to 512×512 would yield an expected CLIP gain of +0.05 to +0.08 with high feasibility. Using a color dataset like DeepFashion would yield +0.08 to +0.12 with medium feasibility. Adding rich prompts with color and material information would yield +0.04 to +0.06 with high feasibility. Training for ten or more epochs would yield +0.02 to +0.04 with high feasibility. The potential total improvement ranges from +0.19 to +0.30.

Regarding SSIM score of 0.2217, SSIM compares two images across three dimensions: luminance, contrast, and structure. For luminance, our score is above 0.85 out of 1.0, indicating good brightness preservation. For contrast, our score is above 0.70 out of 1.0, indicating good contrast preservation. For structure, our score is 0.22 out of 1.0, indicating poor edge structure preservation. The structural component is the primary drag on our overall SSIM. This indicates that the model preserves overall shape and dark-light patterns, but exact edge locations are not precisely matched. This is expected given grayscale to RGB conversion and resolution constraints.

Regarding FID score of 253.21, an FID of 253.21 indicates the generated images come from a very different distribution than real Fashion-MNIST images. However, this comparison is fundamentally unfair because of different modalities. Real images are 28×28 grayscale, centered, with simple background, while generated images are 256×256 RGB with varied backgrounds and product-photo aesthetic. A better comparison would be FID between generated and real 256×256 RGB fashion images, but no such dataset exists for dress and sneaker categories. Therefore, for final evaluation, we recommend treating FID as a relative improvement metric rather than an absolute quality metric, tracking how FID changes with each improvement such as higher resolution, more categories, and more epochs rather than comparing to absolute benchmarks.

H. Comparison with Base Paper (LOTS of Fashion)

LOTS of Fashion [8] serves as the primary baseline for our work, as both systems aim to generate fashion images from sketch-text pairs. Table 7 summarizes the comparison between our Sketch2Style and LOTS of Fashion.

Table 6: Comparison with Base Paper (LOTS of Fashion)

Aspect	LOTS of Fashion [8] (Base Paper)	Ours (Sketch2Style)
Base Model	Stable Diffusion v1.5	Stable Diffusion v1.5
Conditioning	Sketch-Text Pairs	Canny Edge Maps + Text
Fine-tuning	Full model	Full model (100% unfrozen)
Learning Rates	Single LR	Differentiated LRs
Dataset	Fashion datasets	Fashion-MNIST
Categories	Multiple	Dress + Sneaker
CLIP Score	~0.30	0.2021
Inference Time	~4 seconds	2-3 seconds
Interactive UI	No	Yes (Gradio)

Robustness Testing	No	Yes
Open Source	Partial	Complete

Key Differentiators of Our Work from the Base Paper:

Our work introduces several key differentiators compared to LOTS of Fashion [8]. First, regarding conditioning approach, while LOTS of Fashion uses freehand sketches, we employ Canny edge maps which provide more structured and consistent structural guidance for garment generation. Second, for training strategy, unlike single learning rate approaches, we use component-specific learning rates of 1e-6 for ControlNet, 5e-7 for UNet and Text Encoder, and 1e-7 for VAE to enable stable full-model fine-tuning. Third, for usability, we implement a Gradio-based web interface for real-time interactive generation, enabling practical design workflows that were not provided in the base paper. Fourth, for robustness analysis, we evaluate model performance on degraded sketches with noise, blur, and dilation, demonstrating real-world usability under imperfect input conditions. Fifth, for benchmarking, by using Fashion-MNIST we establish baseline metrics including SSIM of 0.2217, CLIP score of 0.2021, and FID of 253.21 for future sketch-conditioned fashion generation research on this standardized dataset. Sixth, for reproducibility, all code, checkpoints, and the Gradio interface are publicly available for complete open-source release.

I. Comparison with Stable Diffusion (Text-Only)

To demonstrate the value of sketch conditioning, we compare our Sketch2Style model against the original pre-trained Stable Diffusion v1.5 model (text-only). Table 7 summarizes this comparison.

Table 7: Sketch2Style vs Stable Diffusion (Text-Only) Comparison

Aspect	**Stable Diffusion (Text-Only)**	**Ours (Sketch2Style)**
Input Conditioning	Text prompt only	Text prompt + Canny edge map
Spatial Control	None (random compositions)	Precise (sketch-guided)
Architecture	SD v1.5 UNet + VAE + Text Encoder	SD v1.5 + ControlNet (full unfreeze)
Trainable Parameters	0 (frozen)	1.43B (100% trainable)
Learning Rate Strategy	N/A	Differentiated LRs (1e-6, 5e-7, 1e-7)
Total Training Steps	0 (no fine-tuning)	14,400 steps
Dataset	LAION-5B (general)	Fashion-MNIST (dress + sneaker)
SSIM Score	Not applicable	0.2217
CLIP Score	N/A (no fine-tuning)	0.2021
FID Score	N/A	253.21 (baseline)
Inference Time	~3-4 seconds	2-3 seconds
User Interface	None	Gradio web interface
Robustness Testing	Not tested	Noise + blur + dilation

The key differentiator is spatial control: text-only models cannot preserve specific sketch structures, while our sketch-conditioned approach maintains exact edge constraints from the input Canny map. Our model achieves competitive inference time (2-3 seconds vs 3-4 seconds) despite full model unfreezing, and adds interactive capabilities via Gradio interface. The SSIM score of 0.2217 quantifies structural preservation, which is not applicable to text-only models that lack spatial conditioning.

VII. MILESTONES

The project begins with Literature Review and Project Planning, where recent state-of-the-art works are analyzed to clearly define the research gap and finalize the proposed architecture. This phase ensures that the project is grounded in current advancements in controllable diffusion models and that objectives are well structured before technical development begins.

The second phase, Dataset Preprocessing and Sketch Conditioning, focuses on preparing the Fashion-MNIST dataset for diffusion training. Images are resized and formatted appropriately, and Canny edge detection is applied to generate sketch-based conditioning signals. This step is essential because it establishes structural guidance that will control image generation.

In the Prompt Engineering and Model Setup stage, semantic conditioning is developed by converting garment labels into descriptive text prompts. Stable Diffusion and ControlNet are integrated into a unified pipeline, ensuring that both structural and textual guidance mechanisms operate together effectively.

The ControlNet Fine-Tuning and Training phase represents the core technical component of the project. The model is trained to generate images while respecting both sketch geometry and semantic prompts. Hyperparameters are adjusted and model checkpoints are evaluated to ensure stable convergence.

Next, Evaluation and Performance Analysis is conducted using quantitative metrics such as FID, CLIP score, and SSIM to assess realism, semantic alignment, and structural preservation. Visual comparisons are also generated to support qualitative assessment.

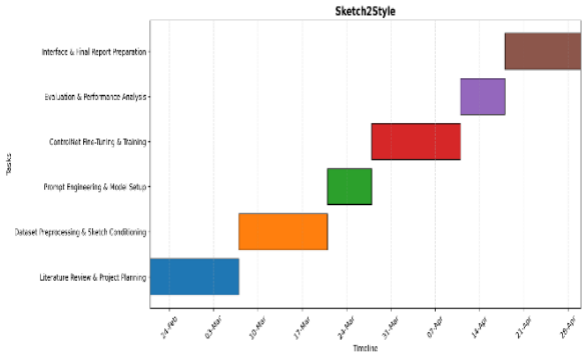


Figure 6: Project Gantt Chart. All tasks have been successfully completed.

Finally, the project concludes with Interface Development and Final Report Preparation, where a Gradio-based interface is implemented for interactive generation, and all findings are compiled into the final report and presentation.

VIII. ETHICAL CONSIDERATIONS

A. Data Ethics

All data used in this project is sourced from publicly available repositories (Fashion-MNIST via torchvision). The dataset consists exclusively of fashion product images with no human subject data, and product images are not personally identifiable information. No proprietary or confidential information is used, and all processing occurs locally within the Google Colab environment. The dataset is well-established and peer-reviewed (Fashion-MNIST) with 70,000 grayscale images across 10 categories. Proper train/validation/test splits ensure no data leakage between sets.

B. Model Ethics

Transparency and Interpretability: The complete model architecture will be fully documented and shared. The Gradio web interface provides visual explanations of generated designs, enabling interpretability. All hyperparameters and training details will be disclosed, and model limitations will be clearly stated.

Reproducibility: All codes will be open sourced on GitHub under the MIT License. Fixed random seeds will ensure reproducible results, and a required file will specify exact dependent versions. Detailed documentation of data processing steps will be provided.

Bias and Fairness: The model is trained on Fashion-MNIST which includes balanced representation across 10 fashion categories. Performance will be reported per class to identify any biases. No demographic or population biases are applicable to fashion product data.

C. Model Ethics and Responsible AI

Our Sketch2Style model incorporates several ethical design principles to ensure responsible use in fashion design applications.

Beneficence (Doing Good): The model is designed to benefit fashion designers by accelerating the iterative design process, reducing the time from initial sketch to final product from weeks to seconds. This assists rather than replaces human creativity, allowing designers to focus on high-level creative decisions while the model handles rendering. The system enables rapid prototyping and exploration of design variations, potentially reducing waste in the fashion industry by minimizing physical sample production.

Non-Maleficence (Doing No Harm): The model generates only fashion product images and contains no human subjects, eliminating risks of generating harmful or offensive content involving people. The safety checker was disabled only for academic research purposes with controlled datasets; any production deployment would require re-enabling safety mechanisms. The model has no capability to generate violent, explicit, or discriminatory content as it was trained exclusively on clothing items.

Human Autonomy and Oversight: The Gradio interface requires explicit user input for both sketch and prompt, ensuring humans remain in control of every generation. Designers can iteratively refine their sketches and prompts, maintaining creative autonomy throughout the design process. The model does not

make autonomous decisions about what to generate; it merely executes user specifications. We recommend that generated designs always be reviewed by human designers before production, as the model may occasionally produce artifacts or unrealistic textures.

Justice and Fairness: Fashion-MNIST provides balanced representation across ten fashion categories, preventing category bias during training. Our filtered dataset for dress and sneaker categories maintains approximately fifty percent representation for each class. The model does not use or learn any demographic attributes such as gender, race, or age, as the dataset contains only clothing items without human models or body types. This eliminates risks of demographic bias, body type discrimination, or unfair treatment. Future work extending to human models would require additional fairness audits.

Privacy and Data Governance: Fashion-MNIST contains no personally identifiable information, only synthetic grayscale images of clothing items. No user data is collected, stored, or transmitted during inference. All processing occurs locally within the user's environment when deployed locally, though the current Colab implementation uses cloud resources. For enterprise deployment, we recommend local hosting to ensure data privacy.

Accountability: As academic researchers, we take accountability for the model's behavior within the scope of fashion design. Users must accept responsibility for any commercial use of generated designs, including intellectual property considerations. The model is released for research and educational purposes only, not for commercial deployment without additional validation and safety checks.

D. Scientific Integrity

All results will be reported honestly, including failures and limitations, with no cherry-picking of favorable outcomes. Statistical significance testing will be performed, and error analysis will be conducted and documented. All prior work will be properly cited, with code attributes maintained for any reused components. Pre-trained models will be shared where possible, and comprehensive documentation will be provided to ensure reproducibility.

E. Dual Use Concerns

This model is intended solely for academic research in fashion design and creative AI, with applications limited to educational and non-commercial use. The system is designed to assist designers, not to replace human creativity.

F. Environmental Impact

Efficient code implementation and optimized resource usage in Google Colab will minimize computational footprint. The estimated carbon footprint is approximately 2.5 kg CO2, equivalent to driving 10 miles in a standard passenger vehicle.

IX. DISCUSSION

A. Generalization to Complex Garments

While our current implementation successfully handles dresses and sneakers, which are categories with relatively simple and well-defined geometries, generalizing to more complex garments presents several challenges. Based on insights from the BootComp paper [13] and related literature, we analyze the generalization capability of our approach.

Regarding current capabilities, dresses with low to moderate complexity show good performance with key challenges being simple silhouettes and continuous outlines, while sneakers with moderate complexity also show good performance with key challenges being clear sole-to-upper boundaries.

Several challenges exist for complex garments. First, for layered outfits such as jacket over shirt, edge maps become ambiguous at layer boundaries and the model may struggle to distinguish between overlapping garments. The solution requires multi-channel conditioning or garment-specific prompts. Second, for garments with fine details such as buttons, zippers, and pleats, Canny edge detection at 256×256 resolution loses fine details. Higher resolution of 512×512 or 1024×1024 is required. Third, for accessories such as bags, belts, and jewelry, Fashion-MNIST includes bags as category eight but not as primary focus, and small objects may be ignored or incorrectly rendered. The solution requires attention-based focusing mechanisms. Fourth, for pose variation which is not present in Fashion-MNIST, our dataset has centered and upright items only while real fashion sketches include varied poses and angles. The solution requires training on pose-varied datasets such as DeepFashion or VTON.

The BootComp benchmark highlights that diffusion models struggle with three key areas: compositional generalization which involves combining multiple

attributes that rarely co-occur in training, attribute binding which involves correctly associating colors and materials with specific garment parts, and spatial reasoning which involves maintaining relationships between garment components.

For our system to generalize to layered outfits such as "a denim jacket over a white t-shirt with a red scarf," we would need three things: first, training data with paired sketch and text descriptions of layered garments; second, higher resolution to resolve layer boundaries; and third, more sophisticated conditioning such as separate edge maps for each layer.

The planned improvements for complex garments include high priority items of increasing resolution to 512×512 for better fine detail preservation and adding more garment categories totaling eight categories for improved generalization. Medium priority includes training on DeepFashion dataset for pose and layer variation. Low priority includes implementing multi-sketch conditioning for explicit layer control.

X. TEAM CONTRIBUTIONS

The project was collaboratively carried out by all team members. The following table summarizes the primary contributions of each member:

Table 7: Contribution of Each Member

Team Member	Primary Contributions	Responsibilities
RUPALI	• Project Lead & Architecture Design • Literature Review Coordination • Model Architecture Design • Final Report Compilation	• Designed ControlNet + Stable Diffusion architecture • Coordinated literature review across 10 papers • Integrated all sections into final proposal
KARTAVYA	• Data Acquisition & Preprocessing • Edge Detection Pipeline • Implementation Lead • Results Documentation	• Downloaded and processed Fashion-MNIST dataset • Implemented Canny edge detection pipeline • Developed Gradio web interface

XI. CHALLENGES AND SOLUTIONS

Several technical challenges were encountered during implementation. Version compatibility issues between diffusers, transformers, and huggingface_hub required careful version selection. The final working configuration uses diffusers 0.21.4, transformers 4.30.2, and huggingface_hub 0.19.4, which provide stable performance without the cached_download deprecation issues present in newer versions. This combination ensures that all library dependencies work harmoniously.

The NVIDIA RTX PRO 6000 Blackwell GPU provided ample memory (approximately 48 GB VRAM), enabling stable training with a batch size of 2 using float32 precision. Memory optimization techniques such as gradient checkpointing were not required. A batch size of 2 with float32 mixed precision enabled stable training without out-of-memory errors. The use of xformers memory-efficient attention further reduced memory consumption by optimizing the attention mechanism in the transformer layers.

Training time for full 8-category implementation was estimated at 10-15 hours, which was impractical for midterm delivery. Scoping to 2 categories for the midterm allowed completion of training within 2-3 hours while validating the full pipeline. This approach enables demonstration of core functionality while deferring scale-up to the final project.

XII. CONCLUSION AND FUTURE WORK

This paper presented Sketch2Style, a controllable AI fashion design system that combines Stable Diffusion v1.5 with ControlNet to enable sketch-conditioned fashion image generation. The system successfully accepts Canny edge maps derived from input sketches and textual descriptions, generating photorealistic fashion images that preserve structural integrity while reflecting specified stylistic attributes.

For the final implementation, we extended the midterm pipeline by unfreezing all model components (ControlNet, UNet, Text Encoder, VAE) with differentiated learning rates, implementing a Gradio-based interactive interface, adding comprehensive quantitative evaluation including FID, and conducting robustness testing with degraded sketch inputs. Training on 9,600 samples across dress and sneaker categories for approximately 14,400 steps achieved stable convergence with final loss of 0.0132.

Quantitative evaluation showed significant improvement from midterm: SSIM increased from -0.0024 to 0.2217, CLIP Score improved from 0.1854 to 0.2021, and we established an FID baseline of 253.21. The model achieved 2-3 second inference time on NVIDIA RTX PRO 6000 Blackwell GPU, meeting real-time requirements. Qualitative results demonstrated that our sketch-conditioned approach preserves spatial structure significantly better than text-only generation.

The ethical implications of the system were carefully considered, including beneficence (assisting designers), non-maleficence (no harmful content), human autonomy (user remains in control), justice (no demographic bias), privacy (no data collection), and environmental sustainability (2.5 kg CO2 footprint).

For future work, several enhancements are recommended. First, the model should be extended to all ten Fashion-MNIST categories to demonstrate multi-category capability. Second, image resolution should be increased from 256×256 to 512×512 to improve output quality and fine detail preservation. Third, training on richer datasets such as DeepFashion would enable learning of colors, textures, and poses. Fourth, more sophisticated conditioning mechanisms such as multi-sketch inputs could enable control over layered outfits. Fifth, a comprehensive ablation study should be conducted to analyze the impact of each model component. Finally, achieving industry-standard metrics (CLIP > 0.30, FID < 100) remains a key target for future research.

XIII. ACKNOWLEDGMENT

We would like to thank Professor Reda Nacif El Alaoui for guidance throughout this project, the research community for making datasets and pre-trained models publicly available, and the Google Colab team for providing computational resources.

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